

Reproducing Timeline Self-Reflection for Temporal Reasoning with LoRA Fine-Tuning

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Abstract—Temporal reasoning, tracking event order, duration, and overlap across a textual context, remains a difficult capability for large language models. This project reproduces the TISER framework proposed by Bazaga et al. [1], which trains a model to generate an explicit structured trace (reasoning, timeline, reflection, answer) before producing its final response. We reproduce this setup using Qwen2.5-7B-Instruct with LoRA [5] on the released TISER data: frozen base weights, LoRA adapters on attention projections ($r=16$, $\alpha=32$), bf16 precision, completion-only loss masking, and consistent Qwen chat-template formatting at both training and inference. On the full in-domain test set our reproduction achieves macro-EM 0.878 / macro-F1 0.949, placing it in the same performance regime as the original paper’s reported macro-EM 0.911 / macro-F1 0.944 for the comparable Qwen2.5-7B + TISER configuration. Beyond reproduction, we propose a *Context–Memory Conflict* probe: we select temporal-QA items the model reliably recalls closed-book (correct with no context at $\geq 3/5$ self-consistency), then deterministically edit the context so the context-implied answer contradicts that memory. Evaluating a 2×2 matrix of {fine-tuned vs. vanilla} $\times$ {TISER prompt vs. standard prompt}, we find that the TISER pipeline raises context-faithfulness from 0.380 to 0.787 (faithful-EM), but the <reflection> step almost never explicitly detects the contradiction: a Claude-based LLM-agent audit of all reflections puts the conflict-naming rate at 4.2%, barely above the 2.3% baseline revise-rate. TISER thus turns the model into a grounded reader, but not a genuine conflict auditor. Order-reversal conflicts remain a blind spot where memorised answers dominate. We also implemented a tennis-domain adaptation extension: on Qwen2.5-0.5B-Instruct, a preliminary tennis-only LoRA adapter improves tennis-test EM from 0.379 to 0.464 on the 224-example test set. The project repository is at GitHub repository.

I. INTRODUCTION

Temporal reasoning is the ability to interpret and manipulate time-related information, including event ordering, temporal overlap, duration, and relations between events. In natural language processing, this capability is central to temporal question answering, narrative understanding, scheduling, and retrieval-augmented generation. Despite the strong general performance of recent large language models (LLMs), temporal reasoning remains fragile: models may identify the relevant events but fail to align them correctly in time, confuse start and end points, or produce answers that are semantically correct but not formatted as expected by exact-match metrics.

The paper *Learning to Reason Over Time: Timeline Self-Reflection for Improved Temporal Reasoning in Language Models* introduces TISER, a framework for improving tem-

poral reasoning through explicit timeline self-reflection [1]. Instead of asking a model to directly output an answer, the model is trained to produce a structured trace containing an initial reasoning step, an explicit timeline of relevant events, a reflection step that checks the reasoning against the timeline, and a final concise answer, delimited by <reasoning>, <timeline>, <reflection>, and <answer> tags.

Our project reproduces the TISER baseline using Qwen2.5-7B-Instruct with LoRA fine-tuning on the released TISER data. We then extend the evaluation with a *Context–Memory Conflict* probe that is not part of the original paper. The extension asks whether the model follows the provided temporal context or its parametric memory when they disagree, and whether the <reflection> step explicitly detects the contradiction. This is relevant because TISER presents reflection as a mechanism for checking reasoning, our probe tests whether it also functions as an auditing mechanism under adversarial conflict.

II. PROBLEM STATEMENT

A. TISER Formulation

We address temporal question answering over a textual or semi-structured temporal context. Each example consists of a question q , a temporal context c , and a gold answer a . The model is trained to generate a structured trace z followed by the final answer:

$$y = (r, t, s, a),$$

where r is the initial reasoning, t is the explicit timeline, s is the self-reflection, and a is the final answer, serialized with XML-like delimiters (<reasoning>, <timeline>, <reflection>, <answer>). During evaluation, only the extracted <answer> span is scored.

The reproduction evaluates on five in-domain splits: TGQA [2], TempReason L2 and L3 [3], and TimeQA easy and hard [4]. Out-of-distribution ToT-semantic examples are reported separately.

B. Context–Memory Conflict Probe

The extension probes whether the model follows a provided context or its memorised knowledge when they disagree. Given a real-entity temporal-QA item:

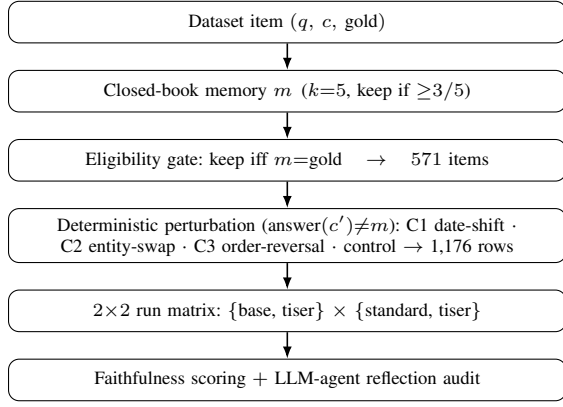


Fig. 1. The inference-only Context-Memory Conflict pipeline: closed-book memory elicitation, an eligibility gate that keeps only confidently-recalled facts, deterministic counterfactual perturbation into conflict classes, a 2×2 model $\times$ prompt run matrix, and faithfulness scoring plus an LLM-agent reflection audit.

- 1) **Memory elicitation.** Remove the temporal context and elicit the model’s closed-book answer m via $k=5$ samples at temperature ≈ 0.7 ; accept the majority answer if it appears in $\geq 3/5$ samples.
- 2) **Eligibility.** Keep only items where m is confident and equals the original gold answer, ensuring a genuine memorised belief.
- 3) **Counterfactual construction.** Deterministically edit the context to produce c' such that $\text{answer}(c') \neq m$.
- 4) **Evaluation.** Run the model on (q, c') and score whether the prediction follows c' or m .

We define: *context-faithfulness* = prediction matches $\text{answer}(c')$; *memorisation* = prediction matches m ; *conflict-detection* = `<reflection>` explicitly names the contradiction (decided by an LLM-agent audit, §III); *silent override* = faithful answer without explicit conflict detection.

We test three hypotheses. **H1:** the TISER prompt and fine-tune make the model more context-faithful under conflict than a plain prompt or the vanilla model. **H2:** when the model is faithful, the `<reflection>` step explicitly names the conflict (otherwise the faithfulness is a *silent override*). **H3:** faithfulness depends on the *type* of conflict (date-shift, entity-swap, or order-reversal). The full extension pipeline is summarised in Fig. 1.

C. Tennis Domain Adaptation for Temporal Reasoning

The second extension is a preliminary tennis-domain adaptation study. It evaluates whether lightweight tennis-only LoRA adaptation improves temporal QA on professional-tennis narratives. Preservation of the original TISER capabilities and mixed replay are left for future work.

III. METHODOLOGY

A. Baseline TISER Reproduction

TISER decomposes temporal reasoning into four stages [1]: reasoning, timeline construction, reflection, and answer generation, all produced autoregressively in a single pass. The target

supervision format teaches the model to externalize temporal structure before answering.

We use the Amazon-released TISER JSONL data. After a 2,048-token length filter, the training set contains 54,488 examples. The test set contains 22,014 examples (19,214 in-domain, 2,800 OOD ToT-semantic).

The base model is Qwen2.5-7B-Instruct, fine-tuned with LoRA [5]: frozen base weights, trainable low-rank adapters on $q/k/v/o$ attention projections, rank $r=16$, scaling $\alpha=32$, dropout 0.05. Training uses bf16 precision without base quantization, 3 epochs, learning rate 2×10^{-4} , cosine schedule, warmup ratio 0.03, effective batch size 16, seed 42, and `paged_adamw_8bit`. Each training sample is converted into a Qwen chat-formatted prompt; the instruction/question/context tokens are masked with label `-100` (completion-only loss). At inference, the model generates the full trace greedily (`max_new_tokens=768`), the `<answer>` span is extracted by regex, and the prediction is normalized using SQuAD-style normalization before computing Exact Match (EM) and token-level F1.

a) *Setup.*: Fine-tuning ran on a single RunPod H200 SXM GPU (bf16 LoRA, no quantization; $54,488 \times 3$ epochs in ~ 6.5 h, ~ 7 samples/s; seed 42) with PyTorch 2.4.1/CUDA 12.4, Transformers 4.46.3, PEFT 0.13.2 and TRL 0.12.2. Probe inference and the full baseline evaluation run on vLLM 0.22.0 (H100 SXM, bf16, greedy); the remaining CPU stages (subset, eligibility, perturbation, scoring, the reflection-audit join and the analysis) run locally.

B. Extension: Context-Memory Conflict Probe

The extension is inference-only (no new fine-tuning); the subject model is the frozen Qwen2.5-7B-Instruct + TISER adapter. Fig. 1 summarises the pipeline.

a) *Subset, memory, eligibility (M0-M2).*: We keep the in-scope real-entity splits (TempReason L2/L3, TimeQA easy/hard; 15,898 items); TGQA is excluded as its synthetic/fictional entities lack the parametric belief needed for a knowledge-conflict probe. For each item the temporal context is deleted and the closed-book answer m is elicited from $k=5$ samples at $T \approx 0.7$, keeping the majority if it appears in $\geq 3/5$ samples. An item is *eligible* iff m is confident and equals the original gold answer, ensuring a genuine memorised belief; this yields 571 eligible items.

b) *Deterministic perturbation (M3).*: Each eligible item is edited into conflict classes: **C1 (date-shift)** moves date ranges so a different entity covers the queried time-point; **C2 (entity-swap)** replaces the answer entity with a same-type distractor; **C3 (order-reversal)** swaps intervals so a before/after relation flips. A *control* arm reorders rows without changing the answer. Each edit enforces the invariant $\text{answer}(c') \neq m$, yielding 1,176 rows (C1 203, C2 520, C3 333, control 120) at 100% per-class construction validity.

c) *Run matrix (M4-M5).*: We cross $\text{model} \in \{\text{vanilla Qwen (base), TISER fine-tuned (tiser)}\}$ with $\text{prompt} \in \{\text{standard direct-answer, TISER 4-tag prompt}\}$; only TISER-prompt cells produce a `<reflection>` trace.

d) *Scoring and reflection audit (M6–M7)*.: M6 scores each prediction for faithful-EM/F1 (matches $\text{answer}(c')$), memorised-EM (matches m) and malformed rate. M7 is a separate post-hoc *LLM-agent reflection audit*: a Claude-based agent reads each `<reflection>` from the TISER-prompt cells and returns a structured verdict—`conflict_raised` (bool), `conflict_kind` $\in$ `{timeline_inconsistency, context_error, context_vs_answer}`, and a short rationale—processed in windows of 100 reflections. It re-generates nothing (it judges the persisted traces) and supersedes a first-pass lexical mention proxy.

C. Tennis Adaptation Dataset and Method

The raw tennis set contains 1,122 temporal QA examples covering before/after relations, first/last events, immediate order, durations, overlap, tournament rounds, and medical timing. Audit and conversion add ids, rule-based categories, prompts, and TISER-compatible fields; duplicate-aware splitting with seed 42 yields 785 train, 113 dev, and 224 test examples.

The tennis evaluator keeps the TISER parser and completion format—`<reasoning>`, `<timeline>`, `<reflection>`, `<answer>`—and adds tennis-specific answer normalization for rankings, minute durations, punctuation, and articles. In addition to base Qwen2.5-0.5B-Instruct standard/TISER-prompt runs, we trained and evaluated a lightweight tennis-only LoRA adapter in the full600 condition. This tennis experiment uses the 0.5B model, not the full Qwen2.5-7B-Instruct TISER reproduction. Original TISER-adapter transfer and mixed tennis+TISER replay were not run.

IV. EXPERIMENTS AND RESULTS

A. Baseline Reproduction

Table I reports our full in-domain test results. Our reproduction achieves macro-EM 0.878 / macro-F1 0.949 across the five in-domain splits. The original paper reports Qwen2.5-7B + TISER at macro-EM 0.911 / macro-F1 0.944 for the comparable in-domain configuration. Our result is in the same performance regime; remaining differences may stem from training infrastructure, data preprocessing, or stochastic variation.

TGQA is the only split below the paper on EM (0.571) yet its F1 is high (0.870): inspecting predictions, $\sim 56\%$ of its well-formed EM=0 cases are unit drops (e.g. “8” vs. “8 years”). Crediting content-correct answers lifts effective TGQA EM to ≈ 0.85 , matching the paper’s 0.845, so the macro-EM gap is largely an exact-match formatting artifact rather than a reasoning gap.

B. Context–Memory Conflict: Run Matrix

Table II reports faithfulness and memorisation rates across all four cells of the run matrix ($n=1,176$ conflict rows per cell).

TABLE I
BASELINE REPRODUCTION ON THE FULL IN-DOMAIN TEST SET, AGAINST THE PAPER’S QWEN2.5-7B + TISER NUMBERS. MACRO IS OVER THE FIVE IN-DOMAIN SPLITS; ToT-SEMANTIC (OOD, EM 0.381) IS EXCLUDED.

Split	n	EM	F1	Paper EM	Paper F1
TGQA	3,316	0.571	0.870	0.845	0.942
TempReason L2	5,397	0.907	0.929	0.855	0.875
TempReason L3	4,426	0.961	0.974	0.915	0.949
TimeQA easy	2,997	0.980	0.989	0.979	0.983
TimeQA hard	3,078	0.970	0.981	0.961	0.972
Macro (5)	19,214	0.878	0.949	0.911	0.944

TABLE II
CONTEXT–MEMORY CONFLICT RUN MATRIX. ALL METRICS ARE RATES OVER $n=1,176$ COUNTERFACTUAL ITEMS. THE *reft-conflict* COLUMN IS THE LLM-AGENT REFLECTION AUDIT, REPORTED ONLY FOR TISER-PROMPT CELLS.

Model	Prompt	faith-EM	faith-F1	mem-EM	malf.	reft-conf.
tiser	tiser	0.787	0.917	0.230	0.002	0.042
tiser	standard	0.574	0.771	0.297	0.000	—
base	tiser	0.561	0.739	0.295	0.003	0.074
base	standard	0.380	0.605	0.332	0.000	—

The TISER prompt raises faithful-EM by +0.213 (95% CI [+0.187, +0.238]) on the fine-tuned model (0.787 vs. 0.574). Fine-tuning raises faithful-EM by +0.225 ([+0.196, +0.256]) under the TISER prompt (0.787 vs. 0.561). Both axes contribute substantially. Item-level bootstrap 95% CIs (2,000 resamples) on the four cells are [.764, .810], [.545, .602], [.532, .590] and [.352, .409], and both head-line gaps are significant under a paired McNemar test (prompt axis $\chi^2=219.9$; model axis $\chi^2=189.9$; both $p<10^{-40}$, $n=1,176$). The base model names conflicts in its reflection more often than the TISER fine-tune (7.4% vs. 4.2%, LLM-agent audit) yet is less faithful, so naming a conflict is not the same as resolving it. Fig. 2 visualises both axes and the per-class breakdown.

C. Per-Conflict-Class Results (Star Cell)

Table III breaks down the star cell (tiser×tiser) by conflict type.

C1 (date-shift) and C2 (entity-swap) are largely solved: the fine-tuned model reliably follows the edited text (faithful-EM ≥ 0.936). C3 (order-reversal) is the blind spot, where faithfulness collapses to 0.366 and memorised answers overtake faithful ones (0.477 vs. 0.366). The audit ties H2 to H3: the model raises the clash almost only on C3 (11.1%, vs. $\leq 2\%$ on C1/C2)—it tends to notice exactly the class it fails. The control arm is a sanity floor, not a faithfulness contest: because $\text{answer}(c') \equiv m$ by construction, its faithful- and memorised-EM coincide (0.900), measuring only whether a benign edit preserved answerability. Control is also a different, L3-heavy subsample (58% TempReason-L3) and is not prompt-independent—the same items score only 0.700 under the standard prompt—so its 0.900 is not proof that the edits are universally clean.

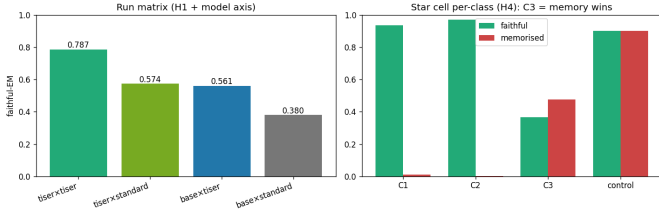


Fig. 2. Left: faithful-EM across the 2×2 run matrix (the H1 prompt axis and the model/SFT axis). Right: per-class faithful vs. memorised EM for the star cell (tiser $\times$ tiser); the C3 order-reversal bar is the only class where memory overtakes the text (H3).

TABLE III

PER-CONFLICT-CLASS RESULTS FOR THE STAR CELL (TISER $\times$ TISER). *agent-refl* IS THE LLM-AGENT CONFLICT-NAMING RATE. FOR *control*, $\text{ANSWER}(c') \equiv m$ BY CONSTRUCTION, SO FAITHFUL- AND MEMORISED-EM COINCIDE.

Class	n	faith-EM	mem-EM	agent-refl	Reading
C1 date-shift	203	0.936	0.010	0.020	follows text
C2 entity-swap	520	0.971	0.002	0.008	follows text
C3 order-reversal	333	0.366	0.477	0.111	memory wins
control	120	0.900	0.900	0.033	sanity floor

D. Tennis Domain Adaptation Results

Table IV reports the main 224-example tennis-test runs. We do not treat smaller smoke conditions as final test results.

The `tennis_only_trace50_smoke_100` run is kept only as a 100-example smoke check (EM 0.330, F1 0.377, one malformed output), not as the main tennis-test result.

V. ANALYSIS

A. Baseline

Our reproduction reaches the same general performance regime as the original TISER paper. EM is brittle for free-form temporal QA, especially when answers differ only by units, punctuation, or symmetric phrasing (the TGQA case above); token-F1 and qualitative inspection are useful complements.

B. Extension

H1 (context-faithfulness) is supported. Both the TISER prompt and TISER fine-tuning independently increase context-faithfulness under conflict. The combined effect raises faithful-EM from the base-standard floor of 0.380 to 0.787.

H2 reveals silent override. The TISER fine-tuned model often follows the edited context (78.7%) but almost never explicitly notices that the context contradicts its memorised belief: an LLM-agent audit of all 1,176 reflections puts the conflict-naming rate at 4.2%, barely above the 2.3% baseline revise-rate measured on normal (non-conflict) data. This rare noticing is driven by conflict *type*, not memory strength—it concentrates on C3, is non-monotonic in the closed-book self-consistency score (peaking at 4/5 rather than 5/5), and even when a clash is flagged the answer still reverts to memory in 23 of the 49 flagged star-cell rows. TISER turns Qwen into a grounded reader, but not a genuine auditor.

TABLE IV

TENNIS TEMPORAL QA RESULTS ON THE 224-EXAMPLE TEST SET WITH QWEN2.5-0.5B-INSTRUCT. DELTAS ARE COMPUTED AGAINST THE BASE STANDARD PROMPT.

Condition	EM	F1	Δ EM	Δ F1	Malf.
Base standard	0.379	0.472	—	—	0
Base + TISER prompt	0.375	0.420	-0.004	-0.052	5
Tennis-only adapter	0.464	0.516	+0.085	+0.045	0

H3 (conflict-type dependence) is supported. Date-shift and entity-swap conflicts are nearly solved (faithful-EM ≥ 0.936), while order-reversal causes a collapse in faithfulness where the model reverts to its memorised event ordering. This is the only conflict class where memorised-EM exceeds faithful-EM.

C. Tennis-Domain Status

TISER prompting alone does not improve the small base model on the full tennis test set: EM changes from 0.379 to 0.375 and F1 drops from 0.472 to 0.420, with five malformed outputs. The tennis-only full600 LoRA adapter improves over the standard base prompt by +0.085 EM and +0.045 F1. This is preliminary because it uses Qwen2.5-0.5B-Instruct and a small synthetic tennis dataset. Original TISER-adapter transfer and mixed tennis+TISER replay remain missing, so we do not report forgetting or preservation of general TISER performance.

D. Caveats

Conflict-naming is decided by a single automated LLM-agent judge; it agrees with the discarded keyword proxy only moderately (Cohen’s $\kappa=0.76$ on the star cell), and no second independent judge or human-calibration round has been run. The eligible set is biased toward facts the model already knows confidently, so 0.787 faithful-EM characterises performance on strong memory conflicts, not a random sample of all temporal QA. All cell scores are single greedy decodes, though we report item-level bootstrap CIs and paired McNemar tests for the headline comparisons. Finally, the reflection text should not be interpreted as mechanistically faithful reasoning.

VI. CONCLUSIONS

Our reproduction of TISER under a single-GPU LoRA setup succeeds, reaching macro-EM 0.878 / macro-F1 0.949 on the full in-domain test set. TISER’s structured trace (reasoning, timeline, reflection, answer) improves temporal QA performance.

The Context–Memory Conflict extension shows that TISER substantially improves context-faithfulness under counterfactual contexts: faithful-EM rises from 0.380 (vanilla baseline) to 0.787 (TISER fine-tune + TISER prompt). However, the reflection stage mostly rubber-stamps the answer rather than explicitly detecting contradictions between the context and memorised knowledge (4.2% conflict-naming rate by the LLM-agent audit vs. a 2.3% baseline revise-rate). Event-ordering conflicts (C3) remain a weakness where the model

reverts to memorised temporal order despite contradictory context.

For the tennis extension, lightweight tennis-only adaptation produces a small but measurable gain on the 224-example tennis test set (EM 0.379 to 0.464). Original TISER-adapter transfer, mixed tennis+TISER replay, and forgetting on the original TISER sample remain future work.

VII. REPRODUCIBILITY

The pipeline is config-driven (`config/config.yaml`, `config/conflict.yaml`) with fixed seed 42. Each base-line and extension run saves `run_meta.json` recording the git SHA, library versions, and resolved configuration. Predictions and per-example EM/F1 scores are saved for error analysis. Extension artifacts are organized under `outputs/conflict/<stage>/`; each pipeline stage (M0 subset, M1 memory, M2 eligibility, M3 conflicts, M4 prompts, M5 generations, M6 scoring, M7 reflection audit, M10 analysis) writes a JSON/JSONL artifact and its own `run_meta.json`. Source code and this report are in the project repository at GitHub repository.

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